6 Fig. 6.1 shows a full-scale diagram of an object O and its image I produced by a converging lens. The lens and its position on the principal axis are not shown.



Fig. 6.1

(a) On Fig. 6.1, draw:

- a single ray to locate the position of the centre of the converging lens
- a line to represent the position of the lens and label the line L.

[2]

(b) Determine the focal length of the lens by drawing another ray on Fig. 6.1.

focal length = ... [2]

(c) The object is moved 2.0 cm closer to the lens.

State **two** changes to the characteristics of the image.

1 .....

2 .....

[2]

[Total: 6]

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